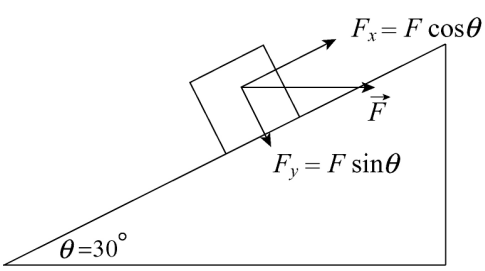
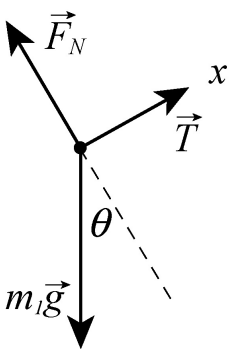
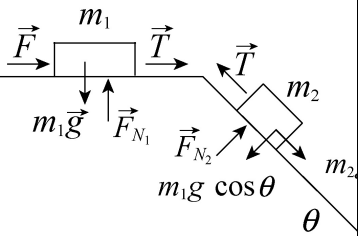

Problems for Ch 5 RW – Forces I
 Addisionele Probleme vir Hfst 5HRW – Kragte I

	9 th Edition	8 th Edition
Questions	1, 4, 9, 10	5, 1, 7, 12
Problems	6, 7, 13, 15, 17, 34, 39, 50, 52, 57, 64, 67, 78, 89, 96	8, 5, 15, 13, 19, 32, 47, 54, 58, 59, 66, 65, 82, 85, 93

6	8	234 N	241 N
7	5	(a) $\vec{F}_2 = (-32.0 \text{ N})\hat{i} - (20.8 \text{ N})\hat{j}$. (b) 38.2N (c) 213°	
13	15	(a) $m_A = 4.0 \text{ kg}$. (b) $m_B = 1.0 \text{ kg}$ (c) $m_C = 4.0 \text{ kg}$ (d) $m_D = 1.0 \text{ kg}$.	
15	13	(a) 108 N (b) 108 N (c) 108 N	
17	19	(a) 42 N (b) 72 N (c) -4.9 m/s^2	
34	32	(a) $F = 651 \text{ N}$ (b) Applying Newton's second law to the y axis (where there is no acceleration), we have $F_N - F \sin \theta - mg \cos \theta = 0$ which yields the normal force $F_N = 1.30 \times 10^3 \text{ N}$	 (a) $F = 566 \text{ N}$. (b) $F_N = 1.13 \times 10^3 \text{ N}$.
39	47	(a) $a = 2.18 \text{ m/s}^2$. (b) $F = 116 \text{ N}$ (c) $a = 21.0 \text{ m/s}^2$	
50	54	(a) 42.0 N. (b) $\Delta x = 0 + \frac{1}{2}at^2 = 0.175$	(a) $T_{BC} = 36.8 \text{ N}$. (b) $\Delta x = 0.191 \text{ m}$.
52	58	$a = 1.7 \text{ m/s}^2$. $v = 5.8 \text{ m/s}$.	$a = 1.3 \text{ m/s}^2$ $v = 5.1 \text{ m/s}$

57	59	 (a) $a = 0.735 \text{ m/s}^2$ (b) The result for a is positive, indicating that the acceleration of block 1 is indeed up the incline and that the acceleration of block 2 is vertically down. (c) $T = 20.8 \text{ N}$
64	66	 (a) $T = 3.0 \text{ N}$. (b) $F = 12.3 \text{ N} \approx 12 \text{ N}$
67	65	$A = 1.63 \text{ m/s}^2$ $T = 81.7 \text{ N}$
78	82	$T = 4.6 \text{ N}$.
89	85	(a) $F = 4.6 \times 10^3 \text{ N}$ (b) $F = 5.8 \times 10^3 \text{ N}$
96	93	$F = 16 \text{ N}$